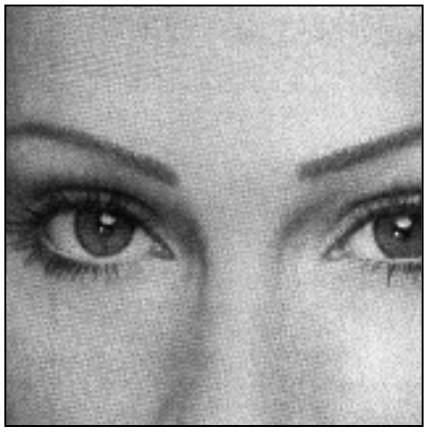
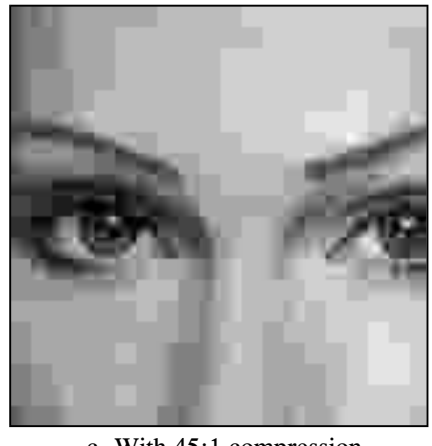


a. Original image



b. With 10:1 compression



c. With 45:1 compression

FIGURE 27-15
Example of JPEG distortion. Figure (a) shows the original image, while (b) and (c) shows restored images using compression ratios of 10:1 and 45:1, respectively. The high compression ratio used in (c) results in each 8×8 pixel group being represented by less than 12 bits.

Why is the DCT better than the Fourier transform for image compression? The main reason is that the DCT has one-half cycle basis functions, i.e., S[0,1] and S[1,0]. As shown in Fig. 27-10, these gently slope from one side of the array to the other. In comparison, the lowest frequencies in the Fourier transform form *one complete cycle*. Images nearly always contain regions where the brightness is gradually changing over a region. Using a basis function that matches this basic pattern allows for better compression.

MPEG

MPEG is a compression standard for digital video sequences, such as used in computer video and digital television networks. In addition, MPEG also provides for the compression of the sound track associated with the video. The name comes from its originating organization, the *Moving Pictures Experts Group*. If you think JPEG is complicated, MPEG is a nightmare! MPEG is something you buy, not try to write yourself. The future of this technology is